MINING DEALER BEST PRACTICE

IMPROVED GPS ANTENNA HARNESS ROUTING IN 794AC COMMAND TRUCKS

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

July 2023
1.10-230724-1

1.0 Introduction

This best practice aims to develop an interim solution to avoid GPS antennas connector (PN: 492-7319) and GPS coaxial harness (PN: 569-9658 LH; 569-9659 RH) damage by falling rocks from the canopy.

The following illustration 1 shows the total downtime associated rocks damaging the harness in the goal post in Quellaveco - 794 autonomous fleet.

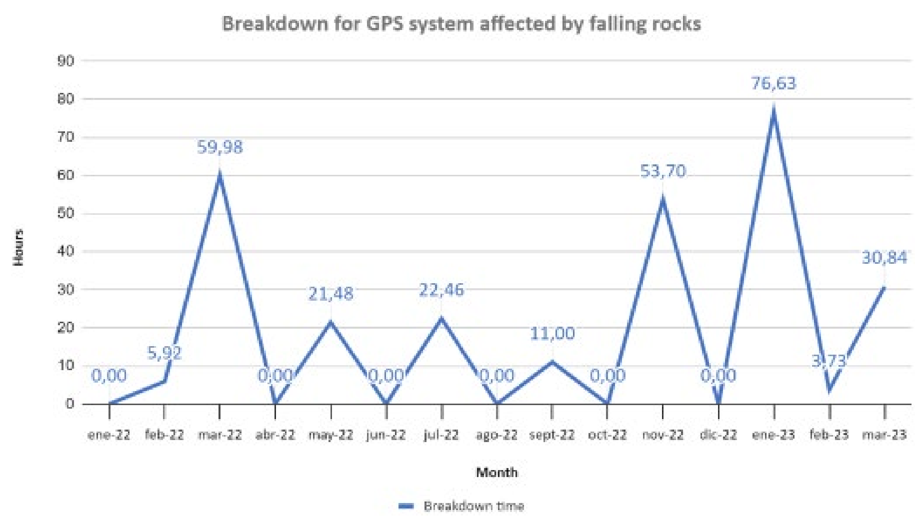

***Illustration 1**: Downtime associated.*

Some reasons for these rocks coming down from the canopy could be related to:

- Suddenly detentions in high-speed conditions.
- Wrong practices during the loading stage.
- Wrong load distribution in the body.

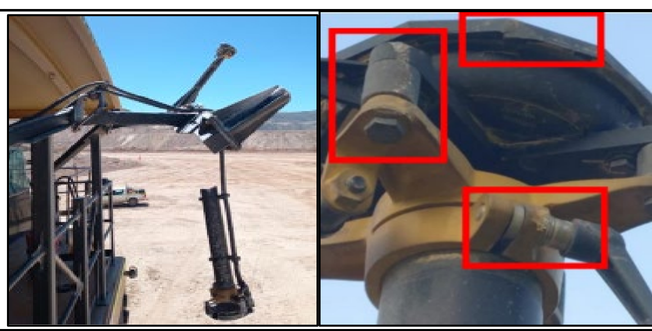	
Damage caused by falling rocks on GPS antennas and harness	GPS coaxial harness damaged or broken by falling rocks

***Illustration 2:** Examples of damage generated in the goal post by rocks falling from the canopy.*

Symptoms about this failure mode are: CMD truck triggers the following events.

- E1252 Low position of the machine.
- E2290 Primary GPS Antenna insufficient precision.
- E2291 Secondary GPS Antenna insufficient precision.
- E1536 GPS Corrections are not received.
- Event 1394 Unacceptable orientation of the machine.

2.0 Best Practice Description

The implementation of this best practice requires modify the GPS (PN: 569-9658 LH; 569-9659 RH) antenna harness routing in the unseen side of the goal post arms.

Routing the coaxial harness away line of fire for falling rocks will avoid the downtime events for troubleshooting (work at height) and parts replacement that will increase the maintenance cost in long term.

The next illustrations show the current design vs the modification proposal.

1

Illustration 3: *Current Design*

2

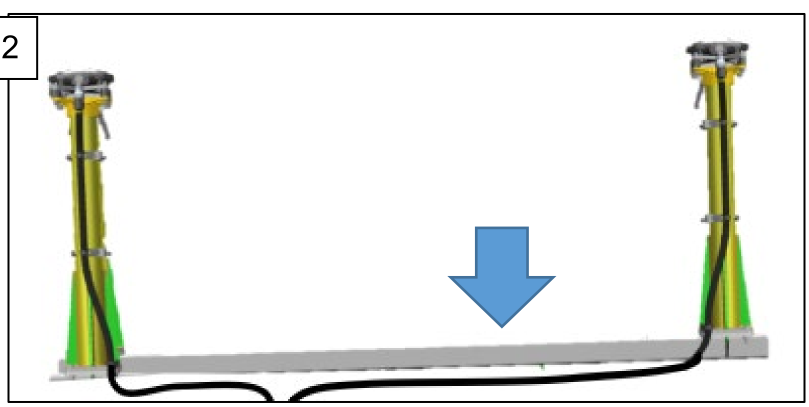

Illustration 4: *Proposal modification*

3.0 Implementation Steps

The current installation of the Master (PN: 569-9658 LH) and Slave GPS (569-9659 RH) antenna harness is as follows:

Illustration 5: *Current arrangement*

The GPS antenna harness is installed on the upper horizontal support of the Goalpost structure.

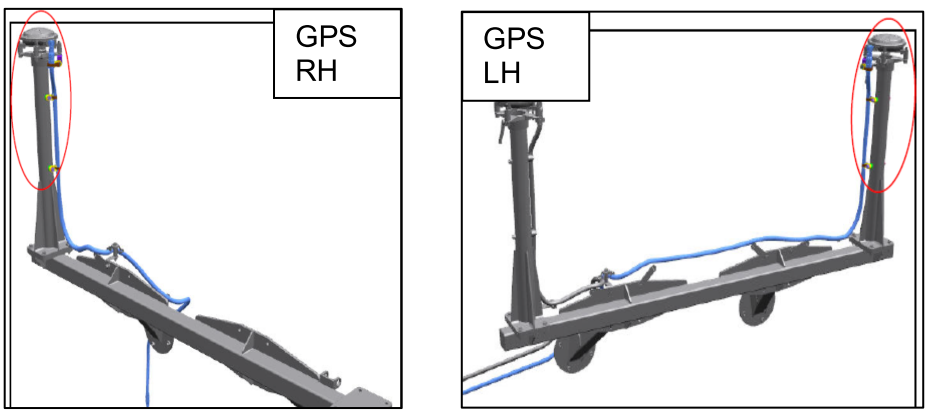

Illustration 6: *Actual routing for RIGHT side and LEFT side coaxial harness*

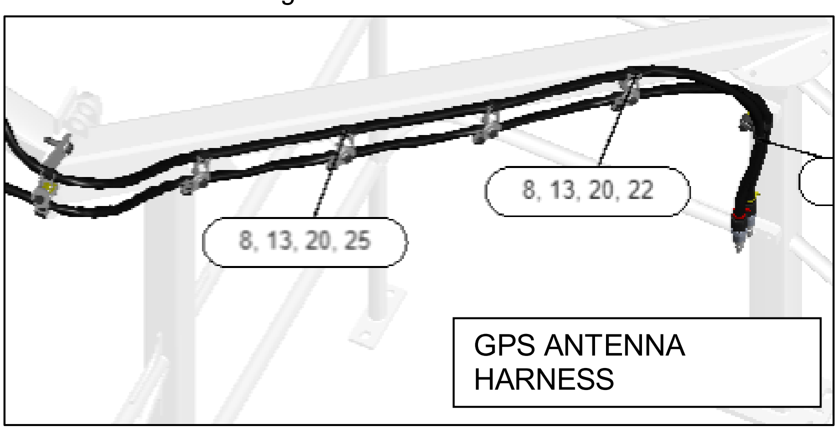

Illustration 7: *Actual routing for both RIGHT and LEFT side coaxial harness upon the platform*

The implementations steps are the following:

a) Loosen the Bracket As (252-9628), rotate the GPS 1(Master PN: 492-7319) and GPS 2(Slave PN: 492-7319) antenna 90° with the antenna connector facing the front.

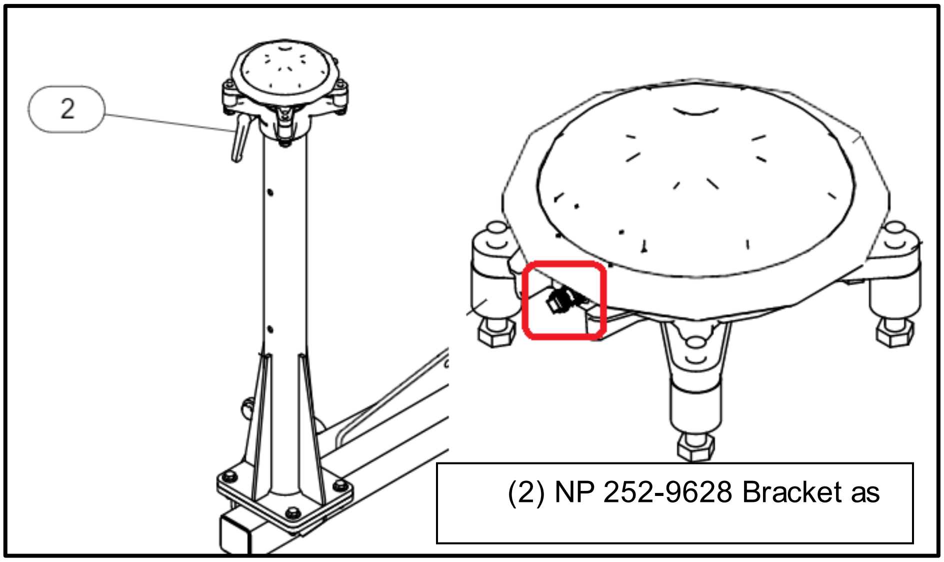

Illustration 8: *Loosen support and rotate both antennas.*

b) The harness of the GPS 1 and GPS 2 antennas are routed through the front part of the vertical support.

- The following NPs are used to fasten the harness:

ITEM	NP	COMPONENT
1	8T-3490	Nut-Weld (M10X1.5-THD)
5	144-4613	Angle AS
7	198-4777	Washer-Hard (11X25X10-MM Thk)
8	204-2281	STRAP-CABLE
9	204-8000	Mounting-Cable Tie
12	344-5675	Nut (M10X1.5-THD)
16	479-2172	Bolt AS
20	8T-4121	Washer-Hard (11X21X2.5-MM Thk)
29	8T-6685	Bolt (M10X1.5X110-MM)

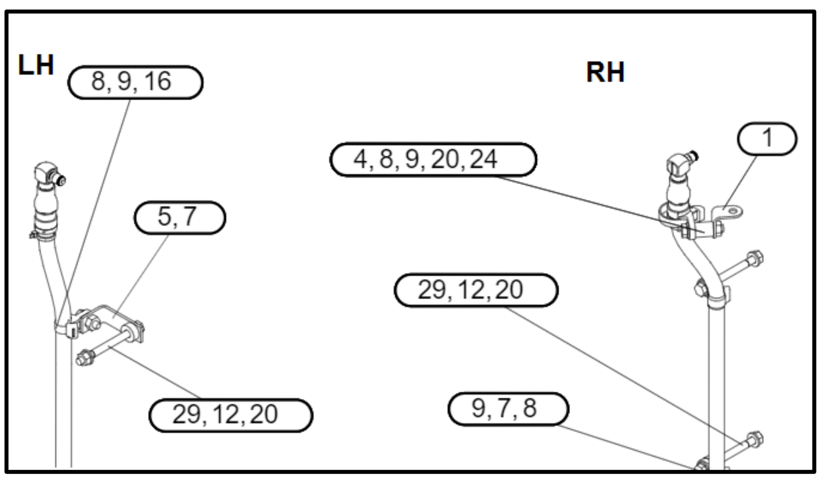

Illustration 9: *Route the outcoming harness for each GPS antenna*

- The fastening of the harness is as follows:

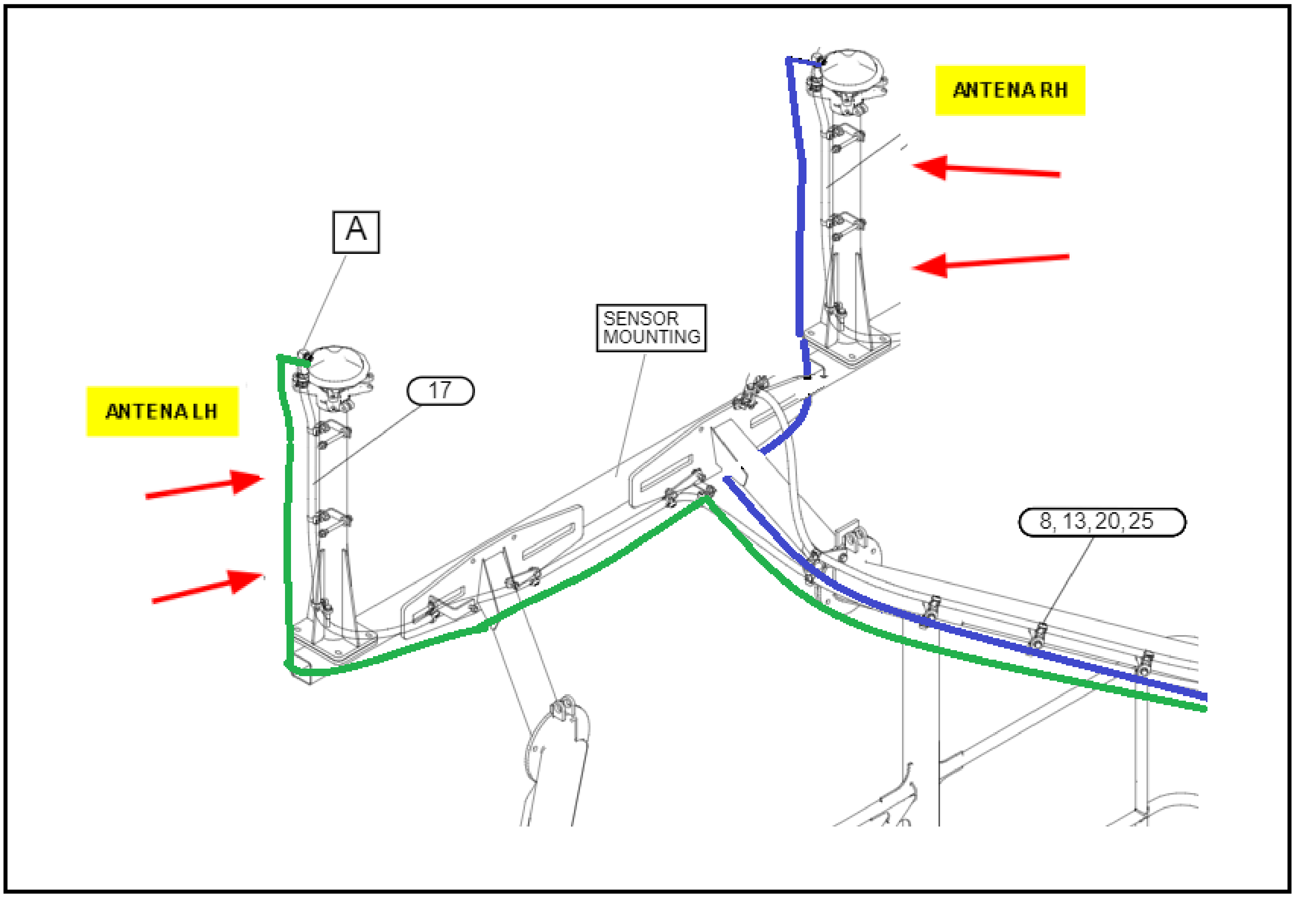

Illustration 10: *Route for coaxial harness*

c) To fasten the harness to the base of the antenna support, the following parts are used on both sides:

ITEM	PN	COMPONENT
8	204-2281	STRAP-CABLE
14	381-2621	Mounting-Cable Tie
20	8T-4121	Washer-Hard (11X21X2.5-MM Thk)
28	8T-6466	Bolt (M10X1.5X60-MM)

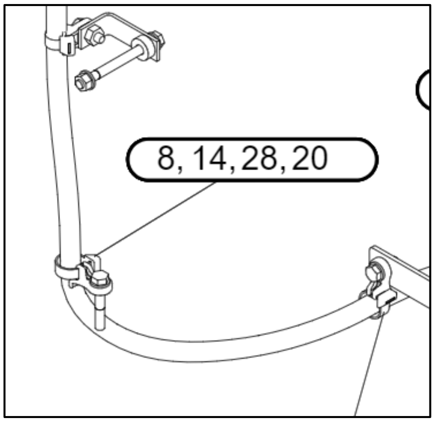

Illustration 11: *Base antenna support hardware*

d) The harnesses of the GPS antennas are attached to the lower part of the horizontal structure.

- The two structural supports (1) are reused to hold the antenna harness through the lower holes.

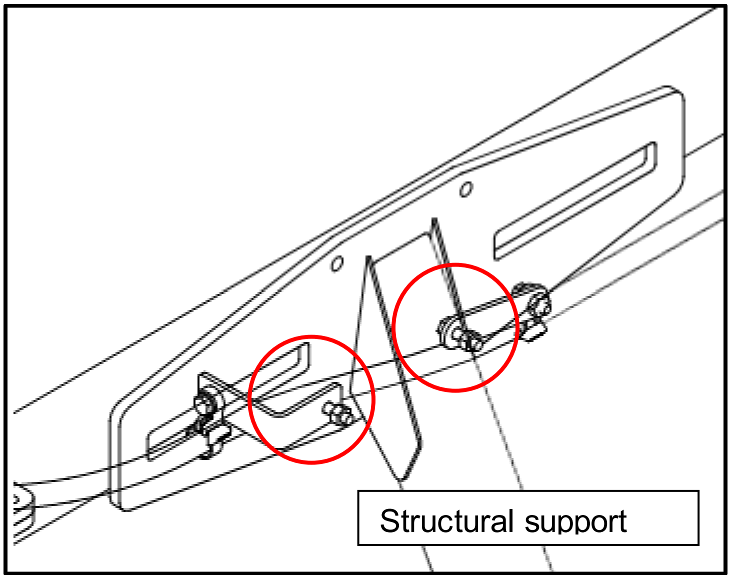

Illustration 12: *LH coaxial harness supported to structure.*

- The harness of the GPS Master antenna is attached through holes A, B and C
.
- The harness of the Slave GPS antenna is attached through hole D.

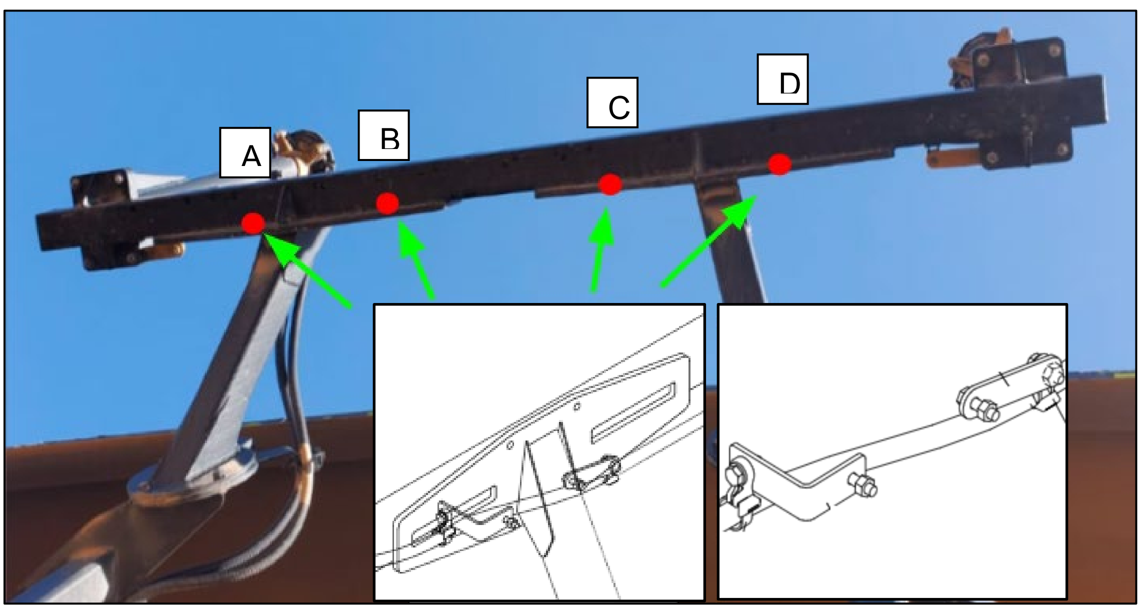

Illustration 13: *Harness routed bellow structure.*

- The following parts are used to fasten the harness:

ITEM	NP	COMPONENT
8	204-2281	STRAP-CABLE
11	322-1358	Bracket AS
11	8T-3490	Nut-Weld (M10X1.5-THD)
12	344-5675	Nut (M10X1.5-THD)
14	381-2621	Mounting-Cable Tie
16	479-2172	Bolt AS
20	8T-4121	Washer-Hard (11X21X2.5-MM Thk)
25	183-7121	Bolt (M10X1.5X35-MM)

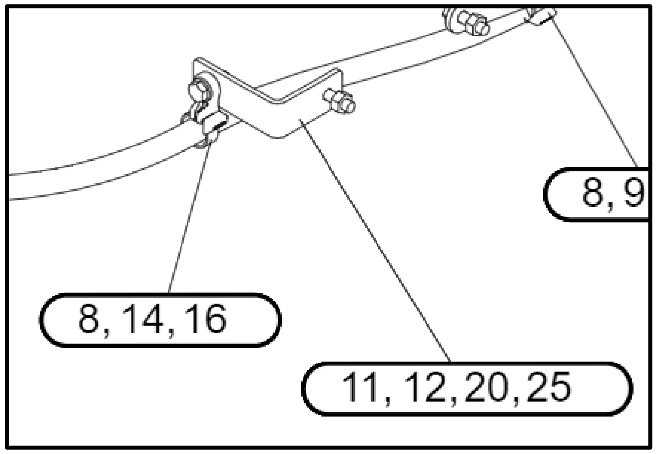

Illustration 14: *Harness fastened by support.*

e) You must weld two bosses (A-B) PN: 294-4550, in the lower part of the structure and attach the harness of the GPS antennas.

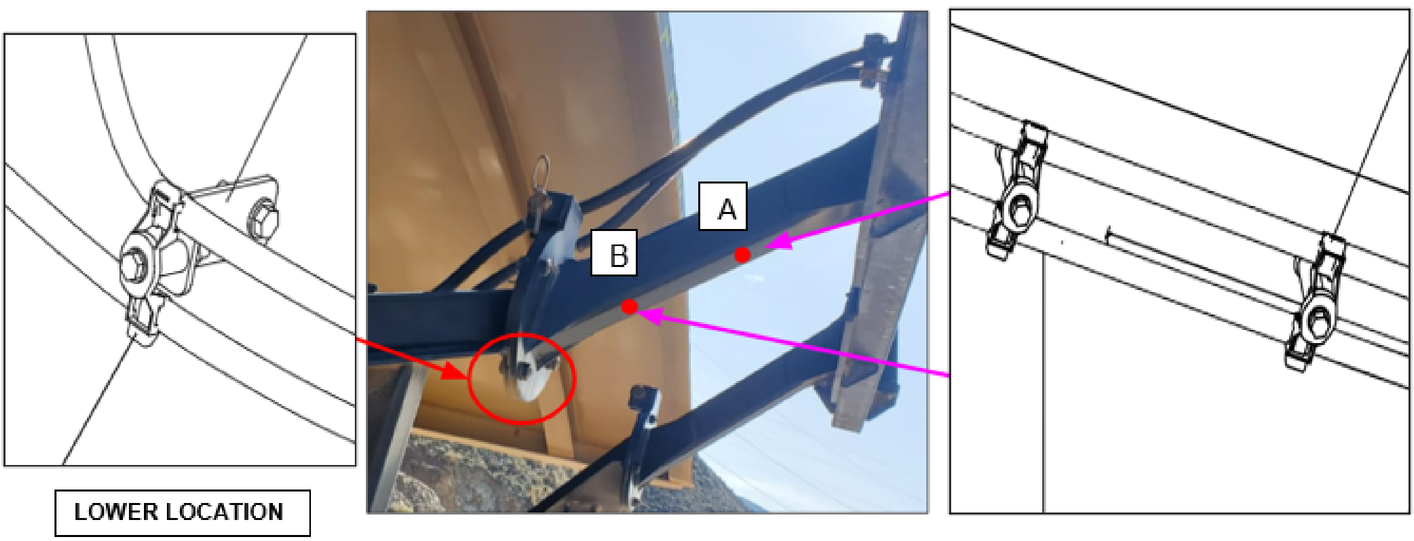

Illustration 15: *Boss welded and support attachments for harness.*

- The following parts are used to fasten the harness:

ITEM	NP	COMPONENT
8	204-2281	STRAP-CABLE
13	345-3753	Mounting-Cable Tie
20	8T-4121	WASHER-HARD
25	183-7121	Bolt (M10X1.5X35-MM)

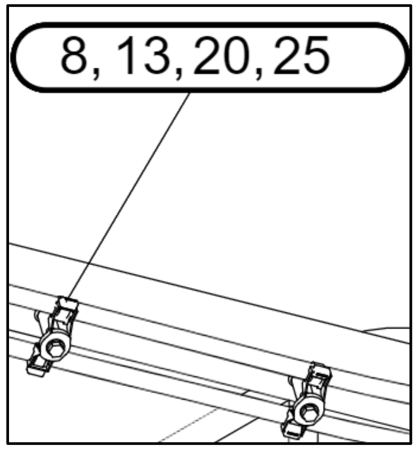

Illustration 16: *Boss welded and support attachments for harness.*

f) Weld 4 bosses (A, B, C and D) PN: 294-4550, at the bottom of the goalpost structure.

g) Install the harness under the structure.

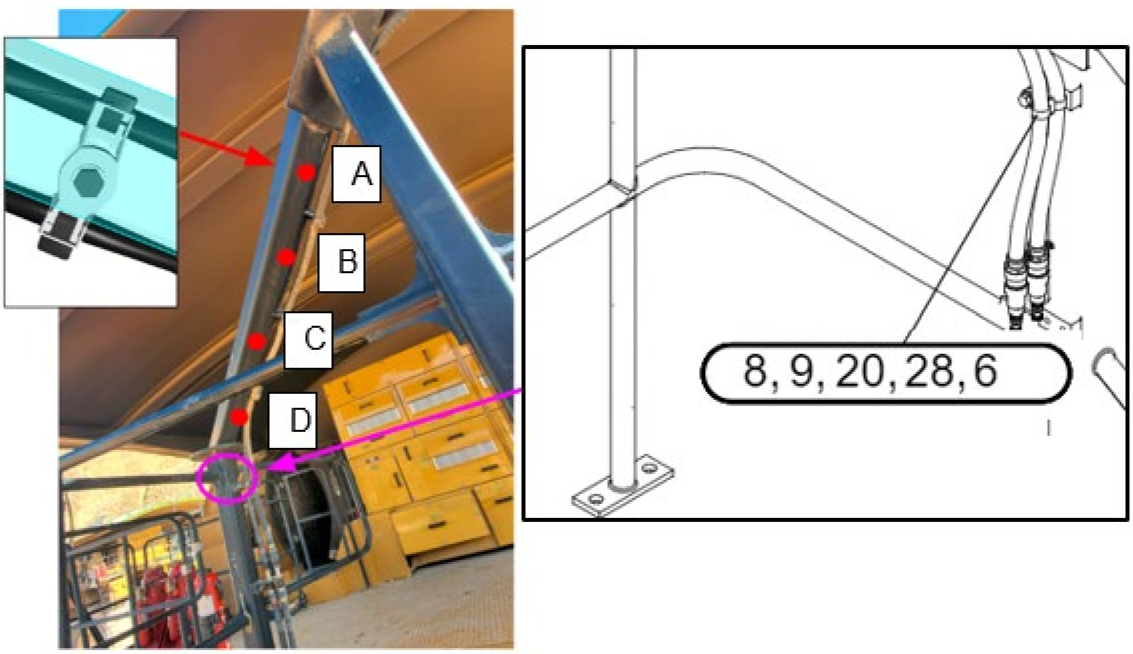

Illustration 17: *Bosses added, and harness support bellow the structure.*

- The following NPs are used to fasten the harness:

ITEM	NP	COMPONENT
8	204-2281	STRAP-CABLE
13	345-3753	Mounting-Cable Tie
20	8T-4121	WASHER-HARD
25	183-7121	Bolt (M10X1.5X35-MM)

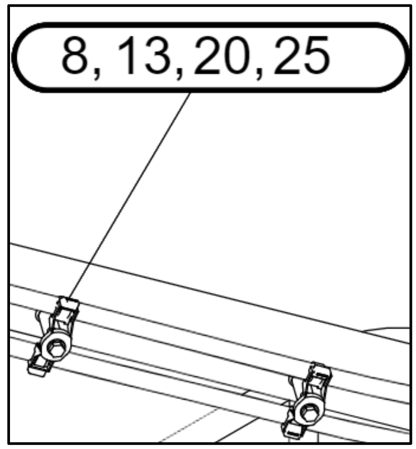

Illustration 18: *Boss welded and support attachments for harness.*

h) Finish the routing in the input connectors to the second GPS antenna harness section (RH-LH).

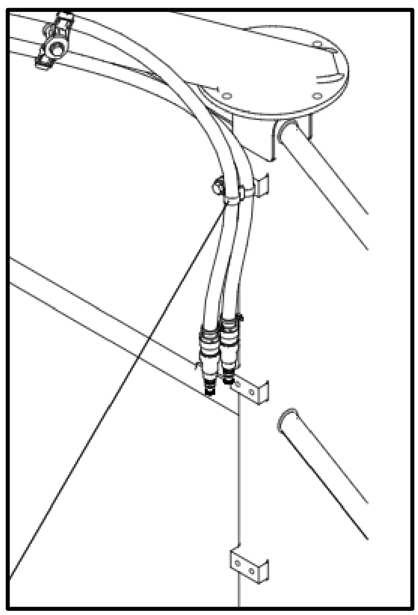

Illustration 19: *Finally route the coaxial harness to the connectors at the platform.*

4.0 Benefits

The benefits of this best practice will be associated with the following topics:

- ✓ Safety: Reduce the time to be exposed to height work.
- ✓ Maintenance: Eliminate the downtime associated with this failure mode, therefore the MTBF and Availability will increase.

5.0 Resources Required

The following parts were used to implement this improvement in the truck:

NP	COMPONENTE	CANTIDAD
127-9323	Bracket AS	2
144-4613	Angle AS	4
144-8344	Spacer(11X19X20-MM Thk)	6
183-7121	Bolt(M10X1.5X35-MM)	7
198-4777	Washer-Hard(11X25X10-MM Thk)	4
204-2281	Strap-Cable	25
204-8000	Mounting-Cable Tie	4
274-7450	Bracket AS	1
322-1358	Bracket AS	4
344-5675	Nut(M10X1.5-THD)	8
345-3753	Mounting-Cable Tie	5
381-2621	Mounting-Cable Tie	6
479-2171	Bolt AS	2
479-2172	Bolt AS	8
8T-3490	Spacer(11X20X40-MM Thk)	8
8T-4121	Washer-Hard(11X21X2.5-MM Thk)	16
8T-4184	Bolt(M12X1.75X45-MM)	1
8T-4223	Washer-Hard(13.5X25.5X3-MM Thk)	1
8T-6466	Bolt(M10X1.5X60-MM)	2
8T-6685	Bolt(M10X1.5X110-MM)	4
294-4550	Boss Weld	6

Resources:

- Manlift
- Welding equipment
- 1 Welder
- 1 electrical technician
- 1 electrical specialist

Installation Time:

- 3.5 hours

Calibrations:

- GAMS

PM to consider:

- 1000hrs
- 2000hrs

Implementation total cost:

6.0 Supporting Attachments / References

794 AC Truck MT5 - SIS 2.0 (cat.com)

7.0 Related Best Practices

None

8.0 Acknowledgements

This best practice was written and implemented by:

Steve A. Tapia Caceres
Electrical trucks support engineer - FERREYROS
steve.tapia@ferreyros.com.pe
phone: (+51) 950 512 464

Brigitte Mamani Concha
Autonomous trucks electrical specialist - FERREYROS
Brigitte.mamani@ferreyros.com.pe
phone: (+51) 951 846 468

Jaisson Chara Taco
Technical Representative / LMT Electric-Drive Mining Trucks - Perú – CATERPILLAR
jaisson.chara@cat.com
mobile: (+51) 915 227 438